



 **CONTACT**

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 1990-05-02

 Christian

 **PROFESSIONAL PROFILE**

02

Self-taught software engineer and systems architect with 20+ years of hands-on experience across software development, industrial systems, networking, automation, data recovery and applied technology.

Builds real-time production monitoring, machine-event and reporting systems, QR traceability flows, mobile operations tooling and recovery-aware workflows.

 **CORE CAPABILITIES**

03

 **ARCHITECTURE**
Systems design and software delivery

 **INDUSTRIAL DATA**
Machine integration and real-time workflows

 **INFRASTRUCTURE**
DevOps, networking and security research

 **APPLIED TOOLING**
Android, reporting and AI-assisted operations

 **TECHNOLOGY STACK**

04

 Python

 Flask

 JavaScript

 SQL

 SQLite

 Linux

 Windows

 Docker

 Nginx

 Git

 Kotlin / Android

 Java / C#

 REST / JSON

 XML

 AI / LLM

 Beckhoff

 Siemens

 MQTT / IoT

SENIOR SOFTWARE ENGINEER

● SYSTEM ONLINE GREEN / TECHNICAL / 2026

DEAN KRUGER

SYSTEMS ARCHITECT • DEVOPS ENGINEER

Engineering practical systems where industrial data, infrastructure and intelligent software meet.

 INDUSTRIAL AUTOMATION

 REAL-TIME SYSTEMS

 AI & ANALYTICS

 CYBERSECURITY

 DEVOPS & INFRASTRUCTURE

 **PROFESSIONAL SUMMARY**

05 / OVERVIEW

Versatile, solutions-driven engineer experienced in developing and deploying connected software for manufacturing and operational environments.

Specializes in real-time production monitoring, machine integration, data collection, QR traceability, mobile applications, reporting and system recovery.

Combines software, infrastructure and operational problem solving to connect factory data with practical decisions.

 **PROFESSIONAL EXPERIENCE**

06 / TIMELINE

 **SOFTWARE ENGINEER / SYSTEMS DEVELOPER**
BTC
Germiston, South Africa

- Developing a manufacturing intelligence platform for production monitoring, machine events, shift reporting and operational dashboards.
- Integrating Maker, Packer and Wrapper production feeds with diagnostics, line status and recovery-aware workflows.
- Building QR traceability and Android operations flows that connect packet records, shifts, machines and operators.
- Designing reporting, KPI, HR / technician and management workflows around a central operational source of truth.

 **FOUNDER / LEAD DEVELOPER**
Legend Systems
Independent practice

- Developing software platforms, operational tools and technical systems for investigations, logistics and business workflows.
- Researching AI-assisted automation, cybersecurity, defensive analysis and practical digital transformation.
- Managing network, server, infrastructure and DevOps work alongside hands-on software development.

 **SOFTWARE & IT CONSULTANT**
Independent Consulting
Web, software and technical support

- Built websites and software solutions while supporting system upgrades, migrations and practical technical problem solving.
- Provided research, documentation and user-facing support for small business and technology projects.

 **SELECTED SYSTEMS**

07 / SYSTEMS

 **Production Intelligence**
Live manufacturing data, machine events, shift reporting and operational dashboards.

 **Line Integration**
Production feeds, diagnostics, line status and recovery-aware workflows.

 **QR Traceability**
Packet records, production context and mobile scanning connected to server-side data.

 **Reporting & Analytics**
Daily, weekly, monthly and yearly production views for practical decisions.

 **Investigations & Logistics**
Secure research, evidence, tracking and logistics concepts.

 **AI-Assisted Operations**
Foundations for research, operational support and reporting workflows.

 **KEY ACHIEVEMENTS**

08 / OUTCOMES

 **MULTI-LINE PRODUCTION PLATFORM**
Connected manufacturing data, events and operational views.

 **REAL-TIME MACHINE MONITORING**
Built workflows around machine status and production context.

 **QR TRACEABILITY WORKFLOW**
Linked packets, shifts, machines and operators through QR flows.

 **OFFLINE RECOVERY ARCHITECTURE**
Designed recovery-aware approaches for disrupted sources and feeds.

 **ANDROID FIELD INTEGRATION**
Connected mobile scanning and operator-facing workflows.

 **OPERATIONAL REPORTING SYSTEMS**
Developed reporting views for practical management decisions.

 **CONTINUOUS LEARNING**

09 / PRACTICE

Continuous hands-on learning across software, Linux, networking, industrial systems, defensive research, mobile tooling, AI-assisted workflows and applied problem solving.